

1|ONELASER

Make Better With One

XRF *Gen2*

The King of RF Desktop Lasers

The Best RF Desktop Laser Just Got Better



www.1laser.com

The most sold **RF desktop laser** in its class.

Exceptional efficiency and category-leading capability — all in a desktop size, built for profitable businesses and passionate makers alike.



As Featured on Fox News "Fox & Friends Weekend" & "Fox Michigan"



Showcased at CES Media Event



Ranked as #1 Desktop RF Laser by industrial reviewer



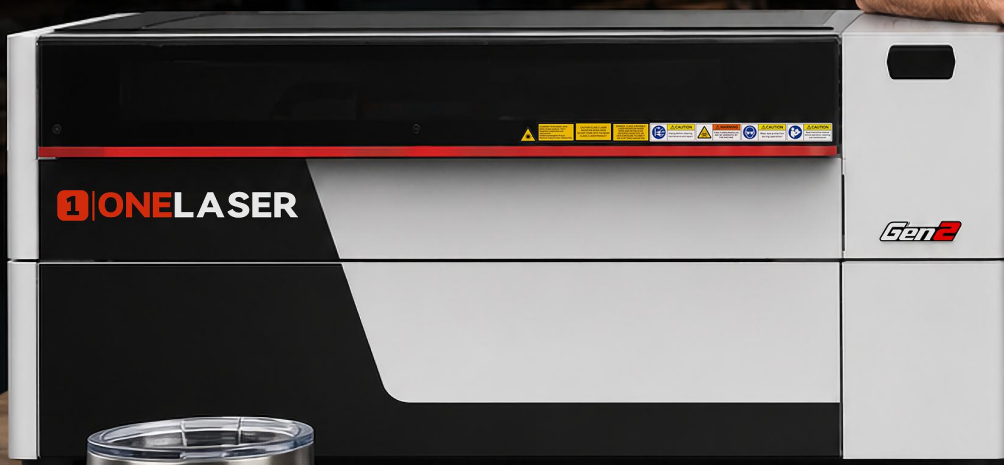
The Best-Selling RF Desktop Laser



Industry-Leading 3-Year Warranty



Powered by OneLaser's Own AI Software



YOUR **WORK** IS ONLY AS **GOOD** AS YOUR LASER.

That's why the best work runs on OneLaser.

OUTDOOR WALL ART

Selling Price: \$149
Net Margin: 76%
Hourly Revenue Output: \$298/hr

EAGLE WALL PANEL

Selling Price: \$119
Net Margin: 82%
Hourly Revenue Output: \$238/hr

TUMBLER

Selling Price: \$39
Net Margin: 72%
Hourly Revenue Output: \$158/hr

WHISKEY GLASS

Selling Price: \$24
Net Margin: 75%
Hourly Revenue Output: \$96/hr

WINE BOTTLE

Selling Price: \$89
Net Margin: 68%
Hourly Revenue Output: \$178/hr

WALNUT SERVING BOARD

Selling Price: \$69
Net Margin: 72%
Hourly Revenue Output: \$207/hr

LEATHER WALLET

Selling Price: \$32
Net Margin: 76%
Hourly Revenue Output: \$128/hr

EATHER PATCH CAP

Selling Price: \$39
Net Margin: 63%
Hourly Revenue Output: \$234/hr

METAL DOG TAGS

Selling Price: \$18 each
Net Margin: 78%
Hourly Revenue Output: \$360/hr

KEYCHAINS

Selling Price: \$16 each
Net Margin: 74%
Hourly Revenue Output: \$384/hr

WOOD/ACRYLIC CHARMS

Selling Price: \$19 each
Net Margin: 75%
Hourly Revenue Output: \$342/hr

Features That Make *XRF* Stands Out.

4K RF Precision

38W / 70W RF tube — air-cooled, long-lasting, photo-quality engraving.

Pinnacle Performance

1,200 mm/s real-world speed at 3.5G acceleration.

IVS (Intelligent Vision System)

High Resolution camera auto-detects Mark points enable print and cut.

FumeWatch Built In

Real-time gas monitoring in a fully enclosed design. Engrave clean, breathe easy.

Endless Engraving, Auto-Fed

Motorized feed + pass-through — oversized jobs, done automatically.

No Alignment Needed

Stable beam path, ready out of the box.

Raised Base, Bigger Reach

Motorized Z handles objects up to 8.5" tall.

Everything at Your Fingertips

Full touchscreen — control, alerts, diagnostics.

Tool-Free Magnetic Optics

Focus lens pops out for cleaning — no tools.

Smart Dual-Mode Air Assist (optional)

Auto-switches cutting and engraving airflow.

Powered by MakerBoost AI

One-click designs and material settings. Zero learning curve.

Instant Fire Protection

Auto-triggered suppression at the first sign of fire (>170°C).

Reengineered Motion System

Low-friction axis for higher precision, longer life, stable speed.

Sealed Tight, Dust Out

Compact, sealed build blocks dust and soot.

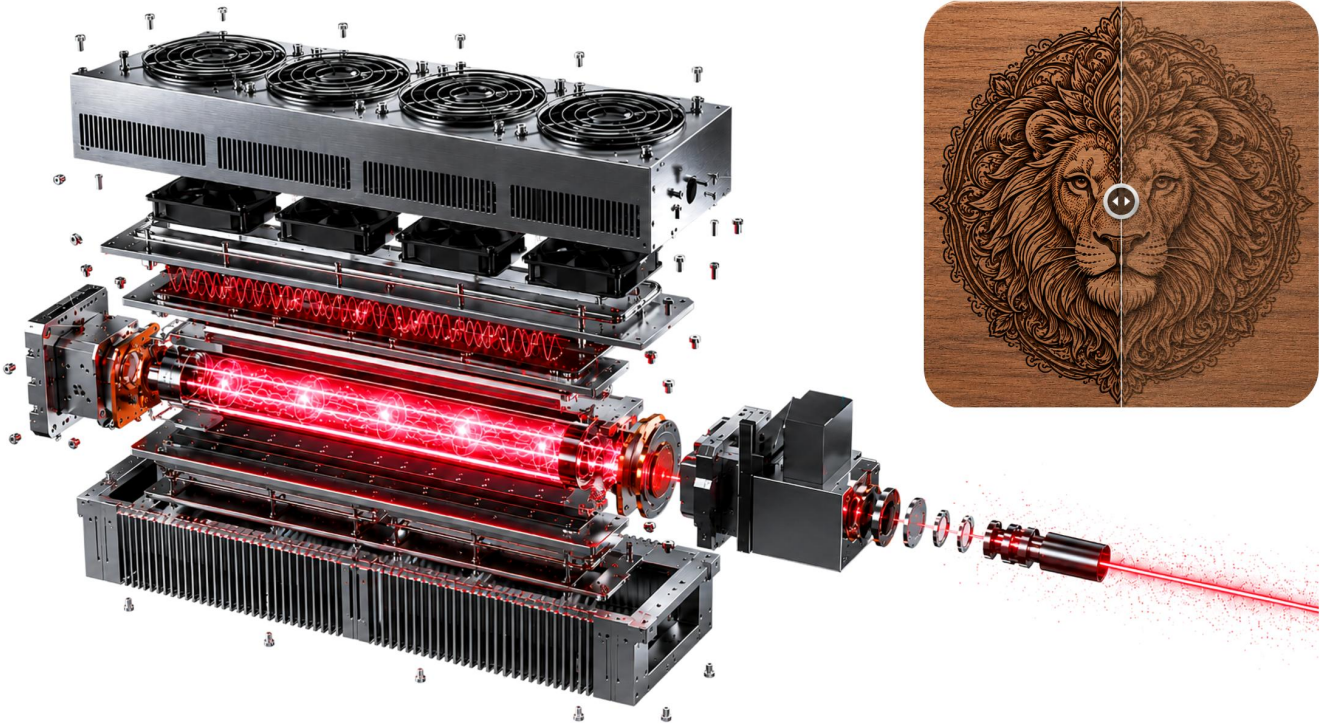
Built Like a Tank for Performance

Rigorously tested — solid, reliable, built to last.



More power. More materials. More possibilities.

The XRF Gen2 now offers a true 70W RF tube—the most powerful RF source in any desktop-class laser. With nearly double the output of the 38W tube and pulse frequencies up to 50 kHz, it delivers ultra-high-DPI flat engraving without sacrificing speed, visibly deeper 3D relief, sharper grayscale and greater cutting capacity.



CLEANER DETAIL

Keep small details sharp—not scorched.

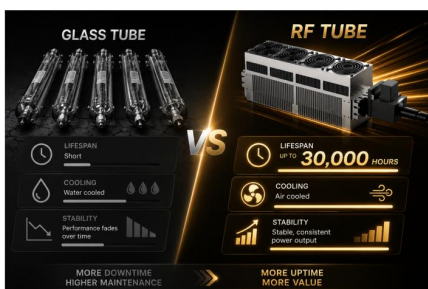
Fine RF pulse control reduces overburn around small text, logos, photos and linework, so every finished piece looks closer to the design on screen.



FASTER RESPONSE

Move faster without leaving quality behind.

RF energy switches on and off quickly, helping XRF hold clean edges and consistent contrast through real engraving work at up to 1,200 mm/s.



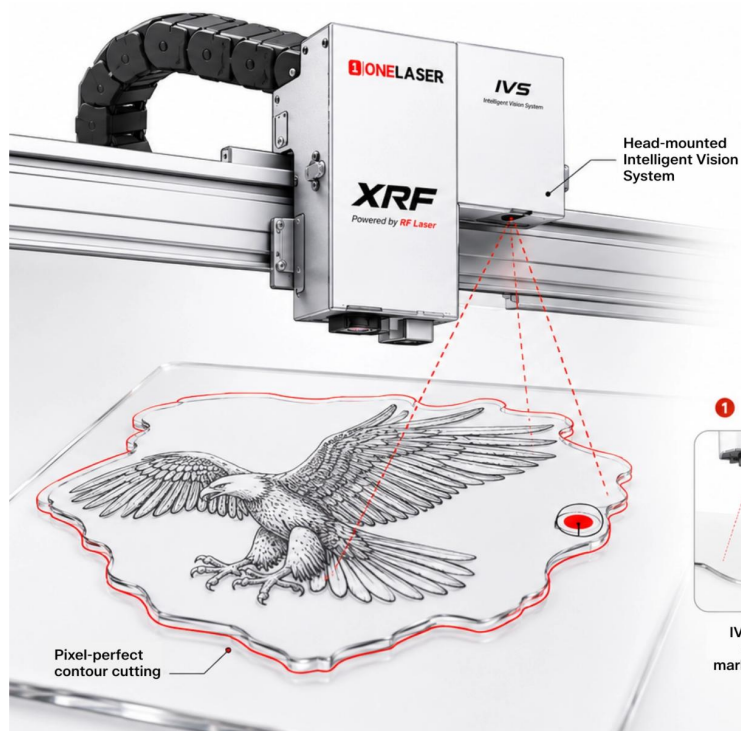
LONGER LIFESPAN

Make more before the source needs attention.

An air-cooled RF source rated for up to 30,000 hours keeps daily production simpler, with no water chiller and more time for the work makers want to ship.

Make every setup decision clearer.

Vision, autofocus and direct machine control reduce the guesswork between an idea and a finished job.



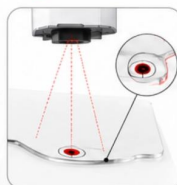
SMART WORKFLOW

Print and cut, made easy.

The head-mounted Intelligent Vision System detects registration marks and compensates position and angle in real time—reducing calibration, waste and rework.

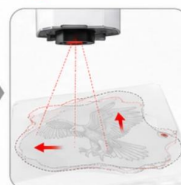


1 DETECT MARK



IVS locates the registration mark on the printed design.

2 COMPENSATE



System calculates position and angle offset in real time.

3 CUT ACCURATELY



Contour is cut with precision despite shifts or misalignment.

Print and cut, made easy.

An integrated vision camera in the laser head performs real-time Mark-point recognition and cut-path compensation — real-time visual registration that delivers precision contour cutting with minimal manual calibration.

Mark detection

Live compensation

Less material waste



Precision in every pixel.

A high-precision camera captures every detail of the bed — grab contours, position with pinpoint alignment, and monitor your job remotely from anywhere.



"One" Touch of Innovation.

More than just one giant button — the intuitive OneLaser touchscreen puts precise head movement, parameter adjustment, and real-time machine status at your fingertips, with clear alarm and fault prompts the moment you need them.

12MP

Top-cover camera · XFocus™ autofocus

Touchscreen

Direct machine control

3 ways

Wi-Fi / USB / Ethernet

Resume

After power returns

A desktop built to grow with your business.

A universal bed, optional height and automatic material handling let the same platform follow a wider product catalog.



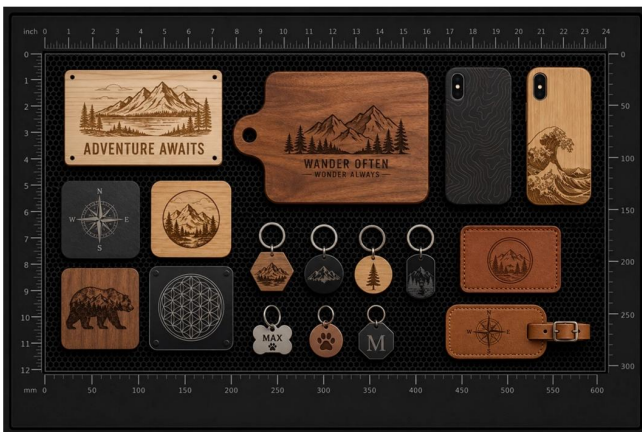
Turn long-format work into one continuous job.

The optional automatic conveyor feeds material in sync with the job, stitching sections into one continuous result for signs, boards and repeating patterns.

Optional Conveyor

Automatic feed

Unlimited project length



One size fits most materials.

The 25.59" x 13.78" (650 x 350 mm) honeycomb table gives you a true 24" x 12" (610 x 305 mm) engraving area — the most universal material size there is, so full sheets load and run without trimming.



Desktop footprint. Big-project capability.

Any object within 24 x 12 x 8.5 inches — paired with the rotary attachment — can be engraved on the XRF: a class-leading working envelope for a desktop machine. The optional Base Boost™ add-on unlocks the full 8.5 inches of height for tall workpieces, 40 oz tumblers and rotary jobs, and opens straight-through access for stock up to 8.5 inches thick.

24 x 12 in

True work area

8.5 in

With optional Riser

Unlimited

Pass-through length

Optional

Rotary workflow

Key Specifications Comparison

Key Metric	ONELASER XRF Gen2	Traditional Desktop Lasers
 Laser Engine	38W/70W RF Metal Tube All-metal sealed, pure air-cooled, with a massive 30,000-hour lifespan	40W Glass Tube Fragile glass, requires external water chiller, with a short lifespan of ~2,000 hours
 Engraving Resolution	Up to 2,000 DPI Ultra-fine 0.07 mm spot size for perfect 3D gradients with zero charring	1,000 DPI Large, coarse spot size, highly prone to heavy scorching and blurry edges
 Max Speed & Motion	1,200 mm/s @ 3.5G Acceleration Features a high-performance industrial DSP motion optimizer	400 - 600 mm/s @ 0.5G High idle time with slow batch turnaround
 Mechanical Rails	Steel Wheel External Rail Guides Large honeycombed bed supporting a 160mm pass-through door	Rubber Roller Wheels Standard fixed small bed size with no pass-through capability
 Workspace Area	24" x 12" (610 x 305 mm) Large honeycombed bed supporting a 160mm pass-through door	18" x 12" (450 x 300 mm) Standard fixed small bed size with no pass-through capability
 Z-Axis Clearance	55 mm to 160 mm Expandable with the Base Boost™ riser, perfect for 40 oz tumblers and thick objects	20 mm to 40 mm Shallow, fixed Z-travel that cannot accommodate rotary attachments
 Vision & Alignment	IVS™ Head-Mounted Camera Seconds-level mark-point scanning with real-time shift and rotation compensation	Lid-Mount Fisheye Camera Severe edge distortion requiring manual positioning and physical jigs
 Active Safety	Built-in Aerosol Extinguisher Suppresses flames automatically; FDA Class 1 fully enclosed with lid safety	Basic Cover Switch Passive shutoff only with no active fire suppression for workshop safety

The World's Best Laser Technology, Shared With Every Maker.

Designed, Supported & Trusted in the **USA**. 
Trusted by **10,000+** Makers Across World.



U.S. Machine Company

A machine partner built for the U.S. market, with accountable delivery, local support, spare parts, users, community, and service you can rely on.

Innovation-Driven Performance

We turn innovation into real productivity: faster workflows, steadier output, and less manual work.

Outcome Guarantee

We stand behind the results you need, from setup and first successful jobs to reliable, measurable production over time.



MEDIA FEATURE: Featured on Fox & Friends Weekend in December 2024.



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Specifications are subject to change without notice.
Product appearance and configuration may vary by model or region.